INFINITE WEIHRAUCH LOOPS: CLOSED-CHOICE FIXED POINTS AND CLOSURE OF NONDETERMINISTIC COMPUTABILITY

VASCO BRATTKA

Editorial scope. One infinite-loop Weihrauch operator/closed-choice fixed-point and nondeterministic-computability closure main group. Injective recursion, countable independent choice, monotonicity, nonclosure and diamond/parallelization comparisons retained as complete proof/support, never separate target counts. Preserve original hypotheses and limitations for partial multivalued problems and Turing degrees.

Allowed standard prerequisites (editorial).

- Author-cited standard foundations. All local definitions, main supporting constructions and complete source bibliography are retained; cited standard foundations remain original references, with no new proof or assumption.

Original abstract. We introduce an operator on problems in Weihrauch complexity, which we call the *infinite loop* or *inverse limit*, and which corresponds to an infinite compositional product. This operation arises naturally whenever one implements algorithms that produce a sequence of results in an infinite loop, using some fixed subroutine. We prove that the corresponding operator is monotone with respect to (strong) Weihrauch reducibility but that it is not a closure operator. One of our findings is that weak Kőnig's lemma is closed under infinite loops, which implies that the class of non-deterministically computable problems is also closed under this operation. Consequently, this class allows for a high degree of flexibility in programming. As our main technical tools, we present an injective version of the recursion theorem and an infinitary version of the so-called independent choice theorem. We also show that, in general, the infinite loop operator is more powerful than the composition of the diamond operator followed by the parallelization operator. However, in many practical scenarios, these compositions yield a result, which coincides with the application of the infinite loop operator. Finally, we discuss the special situation of loops for single-valued problems and for problems on Turing degrees.

1. Introduction

In this article, we explore the power of loops in Weihrauch complexity [BGP21]. In Weihrauch complexity, a multivalued function $f :\subseteq X \rightrightarrows Y$ is seen as a *mathematical problem*, where $f(x) \subseteq Y$ is the set of solutions for an instance $x \in \mathrm{dom}(f)$. Many mathematical problems can be naturally formalized and studied in this way.

We distinguish three different types of loops, which we refer to as *for loops*, *while loops*, and *infinite loops*, respectively. All of these types of loops and further constructions can be captured in Weihrauch complexity by certain operators, as indicated in Figure 1.

The operator of *parallelization* $f \mapsto \widehat{f}$ was introduced by Gherardi and the author [BG11] and was widely studied. Here $\widehat{f} := \mathsf{X}_{i\in\mathbb{N}} f$ stands for the countable parallel application of f. The *diamond operator* $f \mapsto f^\diamond$ was introduced by Neumann and Pauly [NP18], inspired by the concept of generalized Weihrauch reducibility, as introduced by Hirschfeldt and Jockusch [HJ16]. The diamond operator was subsequently characterized by Westrick [Wes21]. It reflects a while loop in the sense that it allows for arbitrarily but finitely many consecutive applications of the problem f in a run of an algorithm.

Date: arxiv:2501.17734v4; CC BY 4.0.

operator	loop
$f^{[*]}$	for loop
$f^{\diamond}$	while loop
f^{∞}	infinite loop
$\widehat{f}$	parallelization

FIGURE 1. Operators and loops.

The number of applications of f is only determined in the course of the computation and not known beforehand. By $f^{[n]}$ we denote the power of a computation, which can use f n–times consecutively, i.e., informally

$$f^{[n]} := \underbrace{f \star ... \star f}_{n-\text{times}}.$$

Here $f \star g$ denotes the *compositional product* of f and g, which was introduced by Pauly and the author [BP18]. It reflects the power of an algorithm, which first uses g and then f. One can then consider the coproduct

$$f^{[*]} := \bigsqcup_{n\in\mathbb{N}} f^{[n]},$$

which reflects the power of a *for loop*, as the number of applications of f has to be determined a priori. Finally, the operator $f \mapsto f^{\infty}$, which we call *infinite loop* or *inverse limit*, is a new one that we introduce here and that intuitively corresponds to an infinite loop that can use the problem f, i.e., informally

$$f^{\infty} := \underbrace{... \star f \star f}_{\text{countably many times}}.$$

More precise definitions follow below. The requirement to use infinite loops occurs often when one computes sequences inductively. One setting where this arises naturally, is in solving initial value problems on their maximal domains of existence [BS25] or also in the proof of the Hahn-Banach theorem [BS26].

One question we study here is which classes of problems are closed under the respective loops. Many classes of problems can be characterized as lower cones of some problem in the Weihrauch lattice. For instance, f is *non-deterministically computable* in the sense of Ziegler [Zie07] if $f \leq_{\mathrm{W}} \mathsf{C}_{2^{\mathbb{N}}}$ holds, i.e., if f is Weihrauch reducible to choice on Cantor space, which in turn is equivalent to weak Kőnig's lemma WKL. Precise definitions can be found below and in [BGP21]. The following table summarizes some of the closure properties of certain classes (a "+" indicates closure, a "−" indicates non-closure): Most of the indicated closure properties can be seen straightforwardly or they follow from known results [BGP21]. In this article we focus on the results regarding *infinite loops* and, in particular, we prove the following theorem that yields the results in the corresponding column of the table.

Theorem 1.1 (Infinite loops)**.** *We obtain:*

(1) $\mathsf{C}^{\infty}_{2^{\mathbb{N}}} \equiv_{\mathrm{W}} \mathsf{C}_{2^{\mathbb{N}}}$, $\mathsf{C}^{\infty}_{\mathbb{N}^{\mathbb{N}}} \equiv_{\mathrm{W}} \mathsf{C}_{\mathbb{N}^{\mathbb{N}}}$,
(2) $\mathsf{C}_{\mathbb{N}} <_{\mathrm{W}} \lim \equiv_{\mathrm{W}} \mathsf{C}^{\infty}_{\mathbb{N}} <_{\mathrm{W}} \lim^{\infty}$.

class of problems	**cone**	**for**	**while**	**infinite**	**parallel**
	f	$f^{[*]}$	$f^{\diamond}$	f^{∞}	$\widehat{f}$
computable	id	+	+	+	+
finite mind-change computable	$\mathsf{C}_{\mathbb{N}}$	+	+	−	−
non-deterministically computable	$\mathsf{C}_{2^{\mathbb{N}}}$	+	+	+	+
limit computable	lim	−	−	−	+
Borel computable	$\mathsf{C}_{\mathbb{N}^{\mathbb{N}}}$	+	+	+	+

FIGURE 2. Classes and closure properties.

Beyond these particular applications, we also seek a more general understanding of the infinite loop operator $f \mapsto f^{\infty}$ and its relationship with other known operators. One interesting question in this context is under which conditions the infinite loop is just the composition of the diamond operator $f \mapsto f^{\diamond}$ followed by the parallelization operator $f \mapsto \widehat{f}$.

Question 1.2. *Characterize (classes of) problems f for which $\widehat{f^{\diamond}} \equiv_{\mathrm{W}} f^{\infty}$ holds!*

Phrased differently, the question is whether infinite loops are more powerful than parallelized while loops, and if so, for which problems? Indeed, it turns out that for many problems f the equivalence stated in the question is actually satisfied. However, this is not always the case.

The structure of this article is as follows. In Section 2, we prove an injective version of the recursion theorem, a key tool that allows us to program infinite loops effectively. Section 3 then establishes precise definitions of the infinite loop and the diamond operator (as well as recalling other standard notations). Section 4 explores basic properties of the infinite loop, including its relation to parallelization and the diamond operator. Section 5 focuses on closure properties of certain choice operations under infinite loops, demonstrating, in particular, that weak Kőnig's lemma is closed under infinite loops. Finally, Section 6 discusses loops in settings where the underlying domain is the set of Turing degrees, illustrating some peculiarities that arise in that context. In particular, we show that, in general, the composition of the parallelization operator and the diamond operator is weaker than the infinite loop operator.

2. Injective recursion theorem

In this section, we state and prove an injective version of the recursion theorem that will be crucial for our analysis of infinite loops. We begin by recalling some necessary preliminaries. They are presented in greater detail in [Bra23]. A function $F :\subseteq \mathbb{N}^{\mathbb{N}} \to \mathbb{N}^{\mathbb{N}}$ is *computable*, if there is some computable monotone word function $f : \mathbb{N}^* \to \mathbb{N}^*$ that *approximates* F in the sense that $F(p) = \sup_{w \sqsubseteq p} f(w)$ holds for all $p \in \mathrm{dom}(F)$. Likewise, F is continuous if and only if an analogous condition holds for an arbitrary monotone word function f. Using this characterization, we can define a representation Φ of the set $\mathcal{C}(\subseteq \mathbb{N}^{\mathbb{N}}, \mathbb{N}^{\mathbb{N}})$ of certain continuous functions $F :\subseteq \mathbb{N}^{\mathbb{N}} \to \mathbb{N}^{\mathbb{N}}$ (with natural domains) by encoding graphs of monotone word functions f into names of F. For details see [Wei87]. Now we can define a computable *universal function*

$$\mathsf{U} :\subseteq \mathbb{N}^{\mathbb{N}} \to \mathbb{N}^{\mathbb{N}}, \langle q, p\rangle \mapsto \Phi_q(p)$$

for all $p, q \in \mathbb{N}^{\mathbb{N}}$ [Wei87, Theorem 3.2.16 (1)]. For simplicity we will write $\mathsf{U}_q = \Phi_q$ in the following. Here $\langle q, p\rangle := q(0)p(0)q(1)p(1)...$ denotes the standard *pairing function* on Baire space. Weihrauch [Wei85, Theorems 3.5, 2.10, Corollary 2.11] (see also [Wei87, Theorem 3.2.16]) proved the following version of the smn-theorem.

Theorem 2.1 (smn)**.** *For every computable (continuous) function $F :\subseteq \mathbb{N}^{\mathbb{N}} \to \mathbb{N}^{\mathbb{N}}$ there exists a computable (continuous) total function $S : \mathbb{N}^{\mathbb{N}} \to \mathbb{N}^{\mathbb{N}}$ such that $\mathsf{U}_{S(q)}(p) = F\langle q, p\rangle$ for all $\langle q, p\rangle \in \mathrm{dom}(F)$.*

Using the smn-theorem one can prove the following uniform version of the recursion theorem along the same lines as the classical recursion theorem. It is an immediate corollary of a more general result due to Kreitz and Weihrauch [KW85, Theorem 3.4] (see also [Wei87, Theorem 3.3.20]).

Theorem 2.2 (Uniform recursion theorem)**.** *There exists a total computable function $T : \mathbb{N}^{\mathbb{N}} \to \mathbb{N}^{\mathbb{N}}$ such that $\mathsf{U}_{T(p)} = \mathsf{U}_{\mathsf{U}_p T(p)}$ for all $p \in \mathbb{N}^{\mathbb{N}}$ such that U_p is total.*

Our goal in this section is to prove a version of the recursion theorem, which simultaneously yields a version of the smn-theorem with a computable injection. We recall that $F :\subseteq \mathbb{N}^{\mathbb{N}} \to \mathbb{N}^{\mathbb{N}}$ is called a *computable injection*, if it is computable and injective and there is a computable function $G :\subseteq \mathbb{N}^{\mathbb{N}} \to \mathbb{N}^{\mathbb{N}}$ such that $G \circ F(p) = p$ for all $p \in \mathrm{dom}(F)$. In the following lemma we prove that program transformations can always be computably turned into computable injections. Intuitively speaking, we can always encode the input of the program transformation as a "comment" into the program text without changing the semantics of the program. This is made formal by the following lemma. For this purpose we assume, without loss of generality, that the descriptions q of functions $\mathsf{U}_q = F$ do allow the digits $0, 1, 2$ as dummy symbols, i.e., adding or removing these digits does not change the meaning of such a name q.

Lemma 2.3 (Injection)**.** *There is a total computable function $I : \mathbb{N}^{\mathbb{N}} \to \mathbb{N}^{\mathbb{N}}$ such that $\mathsf{U}_{I(s)} : \mathbb{N}^{\mathbb{N}} \to \mathbb{N}^{\mathbb{N}}$ is a total computable injection for all $s \in \mathbb{N}^{\mathbb{N}}$ and*

$$\mathsf{U}_{\mathsf{U}_{I(s)}(p)} = \mathsf{U}_{\mathsf{U}_s(p)}$$

for all $s, p \in \mathbb{N}^{\mathbb{N}}$ such that $\mathsf{U}_s(p)$ is defined. In fact, there is a single computable function $L :\subseteq \mathbb{N}^{\mathbb{N}} \to \mathbb{N}^{\mathbb{N}}$ such that $L \circ \mathsf{U}_{I(s)} = \mathrm{id}$ for all $s \in \mathbb{N}^{\mathbb{N}}$.

Proof. If $\mathsf{U}_s(p)$ is defined we can read it as a description of a continuous function $\mathsf{U}_{\mathsf{U}_s(p)}$. We now describe the computation of a total function $F : \mathbb{N}^{\mathbb{N}} \times \mathbb{N}^{\mathbb{N}} \to \mathbb{N}^{\mathbb{N}}$ that yields a new description $F(s, p)$ of the same function, i.e., such that $\mathsf{U}_{\mathsf{U}_s(p)} = \mathsf{U}_{F(s,p)}$ whenever $\mathsf{U}_s(p)$ is defined. We can use the dummy symbols 0 and 1 to encode p, e.g., by adding blocks of the form $10^{p(i)}1$ for $i = 0, 1, 2, ...$ to the encoded list. Simultaneously, we can ensure that there are no other occurrences of the digits 0 and 1 by replacing any of those by the digit 2. This describes how we can compute a new list $F(s, p)$, given $\mathsf{U}_s(p)$. This construction even works if $\mathsf{U}_s(p)$ is undefined (in which case the list will only contain the above blocks of $0, 1$ from a certain point on). This construction ensures that F is total computable and injective in the second component p. In fact, as a function of p it is a computable injection, as we can extract p from the list $F(s, p)$ computably. This extraction is described by a fixed computable function $L :\subseteq \mathbb{N}^{\mathbb{N}} \to \mathbb{N}^{\mathbb{N}}$, not dependent on $s \in \mathbb{N}^{\mathbb{N}}$. By the smn-theorem (Theorem 2.1) there is a total computable $I : \mathbb{N}^{\mathbb{N}} \to \mathbb{N}^{\mathbb{N}}$ such that $\mathsf{U}_{I(s)}(p) = F(s, p)$ for all $s, p \in \mathbb{N}^{\mathbb{N}}$. Altogether, this proves the claim. □

Now we are prepared to prove our injective recursion theorem. We recall that computation on the space $\mathcal{C}(\subseteq \mathbb{N}^{\mathbb{N}}, \mathbb{N}^{\mathbb{N}})$ is understood with respect to the representation Φ.

Theorem 2.4 (Injective recursion theorem)**.** *Let $f :\subseteq \mathcal{C}(\subseteq \mathbb{N}^{\mathbb{N}}, \mathbb{N}^{\mathbb{N}}) \times \mathbb{N}^{\mathbb{N}} \to \mathbb{N}^{\mathbb{N}}$ be a computable function. Then there is a total computable injection $R : \mathbb{N}^{\mathbb{N}} \to \mathbb{N}^{\mathbb{N}}$ such that*

$$\mathsf{U}_{R(q)}(p) = f(R, \langle q, p\rangle)$$

for all $q, p \in \mathbb{N}^{\mathbb{N}}$ such that $f(R, \langle q, p\rangle)$ is defined.

Proof. Let $I : \mathbb{N}^{\mathbb{N}} \to \mathbb{N}^{\mathbb{N}}$ be the computable function from Lemma 2.3. By a double application of the smn-theorem (Theorem 2.1) there is a total computable $S : \mathbb{N}^{\mathbb{N}} \to \mathbb{N}^{\mathbb{N}}$ such that

$$\mathsf{U}_{\mathsf{U}_{S(s)}(q)}(p) = f(\mathsf{U}_{I(s)}, \langle q, p\rangle)$$

for all $s, q, p \in \mathbb{N}^{\mathbb{N}}$, such that the right-hand side exists. Let $t \in \mathbb{N}^{\mathbb{N}}$ be such that $S = \mathsf{U}_t$ and let $T : \mathbb{N}^{\mathbb{N}} \to \mathbb{N}^{\mathbb{N}}$ be the computable function from the uniform recursion theorem (Theorem 2.2). Then $R := \mathsf{U}_{IT(t)}$ is a total computable injection by Lemma 2.3 and we obtain

$$\mathsf{U}_{T(t)} = \mathsf{U}_{\mathsf{U}_t T(t)} = \mathsf{U}_{ST(t)}$$

and hence

$$\begin{aligned} \mathsf{U}_{R(q)}(p) &= \mathsf{U}_{\mathsf{U}_{IT(t)}(q)}(p) = \mathsf{U}_{\mathsf{U}_{T(t)}(q)}(p) = \mathsf{U}_{\mathsf{U}_{ST(t)}(q)}(p) \\ &= f(\mathsf{U}_{IT(t)}, \langle q, p\rangle) = f(R, \langle q, p\rangle) \end{aligned}$$

for all $q, p \in \mathbb{N}^{\mathbb{N}}$ for which $f(R, \langle q, p\rangle)$ is defined. □

3. Infinite loops and diamonds

In this section we provide the exact definition of the infinite loop operation and the diamond operator on problems. We also introduce some concepts from computable analysis and Weihrauch complexity and we refer the reader to [BH21, Wei00] for all concepts that have not been introduced here. We follow the representation based approach to computable analysis and we recall that a *representation* of a space X is a surjective partial map $\delta_X :\subseteq \mathbb{N}^{\mathbb{N}} \to X$. In this case (X, δ_X) is called a *represented space*. A function $F :\subseteq \mathbb{N}^{\mathbb{N}} \to \mathbb{N}^{\mathbb{N}}$ is called a *realizer* of some partial multivalued function $f :\subseteq X \rightrightarrows Y$ on represented spaces (X, δ_X) and (Y, δ_Y), if

$$\delta_Y F(p) \in f\delta_X(p)$$

for all $p \in \mathrm{dom}(f\delta_X)$. In this situation we also write $F \vdash f$. A multivalued map $f :\subseteq X \rightrightarrows Y$ on represented spaces is called a *problem*, if it has a realizer.

We recall that the *composition* $g \circ f :\subseteq X \rightrightarrows Z$ of two problems $f :\subseteq X \rightrightarrows Y$ and $g :\subseteq Y \rightrightarrows Z$ is defined by

$$g \circ f(x) := \{z \in Z : (\exists y \in f(x))\ z \in g(y)\}$$

with $\mathrm{dom}(g \circ f) := \{x \in \mathrm{dom}(f) : f(x) \subseteq \mathrm{dom}(g)\}$.

For simplicity we describe some constructions only for problems $f :\subseteq \mathbb{N}^{\mathbb{N}} \rightrightarrows \mathbb{N}^{\mathbb{N}}$ on Baire space. All definitions can be generalized to arbitrary problems $f :\subseteq X \rightrightarrows Y$ on represented spaces (X, δ_X) and (Y, δ_Y) using standard methods via the *realizer version* of f, defined by $f^{\mathrm{r}} := \delta_Y^{-1} \circ f \circ \delta_X$.

Firstly, we recall the definition of $f \star g$ for problems $f, g :\subseteq \mathbb{N}^{\mathbb{N}} \rightrightarrows \mathbb{N}^{\mathbb{N}}$ from [BGP21]:

$$f \star g := \langle \mathrm{id} \times f\rangle \circ \mathsf{U} \circ \langle \mathrm{id} \times g\rangle.$$

We can define the infinite tupling function $\langle p_0, p_1, p_2, ...\rangle$ and finite tupling functions of higher arity similarly as the pairing function. Now we can define f^∞ in a similar vein as the compositional product. The definition is best understood as an inverse limit construction.

Definition 3.1 (Infinite loops)**.** Let $f :\subseteq \mathbb{N}^{\mathbb{N}} \rightrightarrows \mathbb{N}^{\mathbb{N}}$ be a problem. Then we define the *infinite loop* or *inverse limit* $f^\infty :\subseteq \mathbb{N}^{\mathbb{N}} \rightrightarrows \mathbb{N}^{\mathbb{N}}$ of f by

$$f^\infty(q_0) := \{\langle q_0, q_1, q_2, ...\rangle \in \mathbb{N}^{\mathbb{N}} : (\forall i)\ q_{i+1} \in \mathsf{U} \circ \langle \mathrm{id} \times f\rangle(q_i)\}$$

where $\mathrm{dom}(f^\infty)$ consists of all $q_0 \in \mathbb{N}^{\mathbb{N}}$ such that $A_0 := \{q_0\} \subseteq \mathrm{dom}(\mathsf{U} \circ \langle \mathrm{id} \times f\rangle)$ and $A_{i+1} := \mathsf{U} \circ \langle \mathrm{id} \times f\rangle(A_i) \subseteq \mathrm{dom}(\mathsf{U} \circ \langle \mathrm{id} \times f\rangle)$ for all $i \in \mathbb{N}$. For an arbitrary problem $f :\subseteq X \rightrightarrows Y$ we define $f^\infty := (f^{\mathrm{r}})^\infty$.

That is, the result $\langle q_0, q_1, q_2, ...\rangle$ can be seen as the list of intermediate results that one obtains if the infinite compositional product $... \star f \star f$ is evaluated on input q_0. The domain $\mathrm{dom}(f^\infty)$ of f^∞ is the largest set of inputs for which every infinite run is defined, independently of intermediate choices

We want to rephrase the definition of the diamond operator (see Westrick [Wes21] for a characterization) in similar terms. To this end, the following terminology is useful. We call $\langle q_0, q_1, ..., q_k\rangle$ a *finite run* of the loop on f, if

$$(\forall i < k)\ q_{i+1} \in \mathsf{U} \circ \langle \mathrm{id} \times f\rangle(q_i).$$

Likewise, we define an *infinite run* $\langle q_0, q_1, ...\rangle$ with "$(\forall i)$" instead of "$(\forall i < k)$". Using this terminology, we have

$$f^\infty(q_0) = \{\langle q_0, q_1, q_2, ...\rangle \in \mathbb{N}^\mathbb{N} : \langle q_0, q_1, q_2, ...\rangle \text{ is an infinite run on } f\}.$$

We say that a finite run $\langle q_0, q_1, ..., q_k\rangle$ is *successful*, if

$$q_k(0) = 0 \text{ and } (\forall i < k)\ q_i(0) \neq 0$$

and we say that an infinite run $\langle q_0, q_1, ...\rangle$ is *unsuccessful*, if $(\forall i)\ q_i(0) \neq 0$. Intuitively speaking, we use the condition $q_k(0) = 0$ to indicate that the run has come to a successful end.[1] We say that the run $\langle q_0, q_1, ..., q_k\rangle$ *stalls* if

$$q_k \notin \mathrm{dom}(\mathsf{U} \circ \langle \mathrm{id} \times f\rangle) \text{ and } (\forall i \leq k)\ q_i(0) \neq 0.$$

Now we can define the diamond operator using this terminology as well.

Definition 3.2 (Diamond operator)**.** Let $f :\subseteq \mathbb{N}^\mathbb{N} \rightrightarrows \mathbb{N}^\mathbb{N}$ be a problem. Then we define the *diamond operator* $f^\diamond :\subseteq \mathbb{N}^\mathbb{N} \rightrightarrows \mathbb{N}^\mathbb{N}$ of f by

$$f^\diamond(q_0) := \{q_k \in \mathbb{N}^\mathbb{N} : (\exists q_1, ..., q_{k-1} \in \mathbb{N}^\mathbb{N})\ \langle q_0, ..., q_k\rangle \text{ is a successful finite run on } f\}$$

where $\mathrm{dom}(f^\diamond)$ consists of all $q_0 \in \mathbb{N}^\mathbb{N}$ such that there is no run starting with q_0 that stalls or is infinite and unsuccessful.

Again the definition can be extended to arbitrary problems f using their realizer version f^r. We point out that there is a formal similarity between the definition of the diamond operator and the μ–operator from classical computability theory [Odi89]. Both constructions reflect the power of while loops: the μ–operator does so for single-valued computations on the natural numbers and the diamond operator $f \mapsto f^\diamond$ for arbitrary multivalued problems with f as a subroutine.

Problems can be compared using the tool of (strong) Weihrauch reducibility [BGP21]. By $\mathrm{id} : \mathbb{N}^\mathbb{N} \to \mathbb{N}^\mathbb{N}$ we denote the *identity* on Baire space.

Definition 3.3 (Weihrauch reducibility)**.** Let $f :\subseteq X \rightrightarrows Y$ and $g :\subseteq Z \rightrightarrows W$ be problems. We say that

(1) f is *Weihrauch reducible* to g, in symbols $f \leq_\mathrm{W} g$, if there are computable functions $H, K :\subseteq \mathbb{N}^\mathbb{N} \to \mathbb{N}^\mathbb{N}$ such that $H\langle \mathrm{id}, GK\rangle \vdash f$, whenever $G \vdash g$ holds.
(2) f is *strongly Weihrauch reducible* to g, in symbols $f \leq_\mathrm{sW} g$, if there are computable $H, K :\subseteq \mathbb{N}^\mathbb{N} \to \mathbb{N}^\mathbb{N}$ such that $HGK \vdash f$, whenever $G \vdash g$ holds.

As usual, we denote the corresponding equivalences by $\equiv_\mathrm{W}$ and $\equiv_\mathrm{sW}$, respectively. We recall that a problem f is called *pointed* if $\mathrm{id} \leq_\mathrm{W} f$ holds, i.e., if and only if f has a computable input. The problem $f^\diamond$ is always pointed, as the zero input is a successful run in which no input for f is required.

The main result of Westrick [Wes21] regarding the diamond operator is the following theorem that characterizes the diamond operator as a closure operator that reflects closure under compositional product.

[1] We recall that $0, 1, 2$ are dummy symbols in the names $q \in \mathbb{N}^\mathbb{N}$ of continuous functions U_q. Hence we can use, for instance $q(0) = 0$ or $q(0) = 1$ to indicate success without interference with the meaning of the function U_q.

Theorem 3.4 (Westrick 2021)**.** *For each pointed problem f we have*

$$f^\diamond \equiv_{\mathrm{W}} \min_{\leq_{\mathrm{W}}}\{g : f \leq_{\mathrm{W}} g \star g \leq_{\mathrm{W}} g\}.$$

It is clear that *while loops* can be used to simulate *for loops*. The success condition used in a while loop can simply be that a given number of runs of the loop is performed.

Proposition 3.5. *$f^{[*]} \leq_{\mathrm{sW}} f^\diamond$ for all problems f.*

We close this section with mentioning a number of standard problems that we are going to use in the following (see [BGP21] for more precise definitions). By C_X we denote the choice problem of a computable metric space X, which is defined by $\mathsf{C}_X :\subseteq \mathcal{A}_-(X) \rightrightarrows X, A \mapsto A$, where $\mathcal{A}_-(X)$ denotes the set of closed subsets of X given by negative information. By UC_X we denote the *unique choice problem*, which is the restriction of C_X to singletons. The problem $\mathsf{LLPO} := \mathsf{C}_2 = \mathsf{C}_{\{0,1\}}$ is also known as *lesser limited problem of omniscience*. The problem $\mathsf{LPO} : \mathbb{N}^{\mathbb{N}} \to \{0,1\}$ is simply the characteristic function of $\{\widehat{0}\}$, where $\widehat{0}$ denotes the constant zero sequence. By $\lim_X :\subseteq X^{\mathbb{N}} \to X$ we denote the usual *limit map* of a metric space X, where $\lim := \lim_{\mathbb{N}^{\mathbb{N}}}$ stands for the limit of Baire space. By $\mathsf{J} : \mathbb{N}^{\mathbb{N}} \to \mathbb{N}^{\mathbb{N}}, p \mapsto p'$ we denote the *Turing jump operator*. The problem WKL stands for *weak Kőnig's lemma* and it is the problem $\mathsf{WKL} :\subseteq \mathrm{Tr}_2 \rightrightarrows 2^{\mathbb{N}}, T \mapsto [T]$ that maps every infinite binary tree T to the set of its infinite paths. The following proposition summarizes some well-known results about some of these problems [BG11, BdBP12, BGP21].

Proposition 3.6. *We obtain*

(1) $\widehat{\mathsf{LPO}} \equiv_{\mathrm{sW}} \widehat{\mathsf{C}_{\mathbb{N}}} \equiv_{\mathrm{sW}} \lim \equiv_{\mathrm{sW}} \mathsf{J}$*,*
(2) $\mathsf{C}_{\mathbb{N}}^\diamond \equiv_{\mathrm{sW}} \mathsf{C}_{\mathbb{N}} \equiv_{\mathrm{sW}} \lim_{\mathbb{N}}$*, and*
(3) $\widehat{\mathsf{LLPO}} \equiv_{\mathrm{sW}} \mathsf{C}_{2^{\mathbb{N}}} \equiv_{\mathrm{sW}} \mathsf{WKL}$*.*

4. Basic properties of infinite loops

In this section we prove some basic properties of the infinite loop operation. Programming with infinite loops is not so straight-forward because the "program" to which U is applied in each loop needs to be inherited from the previous loop. This is exactly what can be achieved with the injective recursion theorem. We work this out in technical detail in some of our proofs, but leave the technical details to the reader for most of the others.

We start by showing that $f \mapsto f^\infty$ is actually an operation on (strong) Weihrauch degrees. In fact, the problems f^∞ are all *cylinders* (i.e., $f^\infty \equiv_{\mathrm{sW}} \mathrm{id} \times f^\infty$), hence we always get strong Weihrauch reductions. We write $g \sqsubseteq f$ for two problem $f, g :\subseteq \mathbb{N}^{\mathbb{N}} \rightrightarrows \mathbb{N}^{\mathbb{N}}$ if $\mathrm{dom}(f) \subseteq \mathrm{dom}(g)$ and $g(p) \subseteq f(p)$ for all $p \in \mathrm{dom}(f)$.

Proposition 4.1 (Monotonicity of infinite loops)**.** *$f \leq_{\mathrm{W}} g \Longrightarrow f^\infty \leq_{\mathrm{sW}} g^\infty$ holds for all problems f, g.*

Proof. Without loss of generality, we can assume that $f, g :\subseteq \mathbb{N}^{\mathbb{N}} \rightrightarrows \mathbb{N}^{\mathbb{N}}$ are problems on Baire space. Let $f \leq_{\mathrm{W}} g$. Then there are computable functions $H, K :\subseteq \mathbb{N}^{\mathbb{N}} \to \mathbb{N}^{\mathbb{N}}$ such that $H \circ \langle \mathrm{id}, gK\rangle \sqsubseteq f$. Let $K_2 :\subseteq \mathbb{N}^{\mathbb{N}} \to \mathbb{N}^{\mathbb{N}}$ be the computable function with $K_2\langle q, p\rangle := K(p)$. By the injective recursion theorem (Theorem 2.4) there exists a total computable injection $K_1 : \mathbb{N}^{\mathbb{N}} \to \mathbb{N}^{\mathbb{N}}$ such that

$$\mathsf{U}_{K_1\langle q,p\rangle}(r) = \langle K_1, K_2\rangle \circ \mathsf{U}_q \circ H\langle p, r\rangle$$

for all $q,p,r\in\mathbb{N}^{\mathbb{N}}$ such that the right-hand side exists. That K_1 is a computable injection means that there is a computable function $H_1:\subseteq\mathbb{N}^{\mathbb{N}}\to\mathbb{N}^{\mathbb{N}}$ such that $H_1\circ K_1\langle q,p\rangle=\langle q,p\rangle$ for all $\langle q,p\rangle\in\mathbb{N}^{\mathbb{N}}$ and we obtain

$$\begin{aligned}\mathsf{U}\circ\langle\mathrm{id}\times g\rangle\circ\langle K_1,K_2\rangle\langle q,p\rangle &= \mathsf{U}_{K_1\langle q,p\rangle}(gK(p))\\ &= \langle K_1,K_2\rangle\circ\mathsf{U}_q\circ H\langle p,gK(p)\rangle\\ &\subseteq \langle K_1,K_2\rangle\circ\mathsf{U}\circ\langle\mathrm{id}\times f\rangle\langle q,p\rangle.\end{aligned}$$

Let $H_2:\subseteq\mathbb{N}^{\mathbb{N}}\to\mathbb{N}^{\mathbb{N}}$ be the computable function with

$$H_2\langle\langle q_0,p_0\rangle,\langle q_1,p_1\rangle,...\rangle:=\langle H_1(q_0),H_1(q_1),...\rangle.$$

Then we obtain

$$H_2\circ g^\infty\circ\langle K_1,K_2\rangle\sqsubseteq H_2\circ\widehat{\langle K_1,K_2\rangle}\circ f^\infty=f^\infty$$

and thus $f^\infty\leq_{\mathrm{sW}}g^\infty$. □

With the next result we want to establish some facts on the relation between the diamond operator and infinite loops. In this case we leave the reasoning informal and do not work out the technical details of the application of the inverse recursion theorem, as it would be very technical and block the view on the essential ideas.

Proposition 4.2 (Parallelization, infinite loops and diamonds)**.** *For arbitrary problems f we obtain:*

(1) $\widehat{f^\infty}\equiv_{\mathrm{sW}}f^\infty$,
(2) $f^\diamond\leq_{\mathrm{sW}}f^\infty$, *if f is pointed,*
(3) $\widehat{f^\diamond}\leq_{\mathrm{sW}}f^\infty$, *if f is pointed.*

Proof. (1) We can obtain $f^\infty(p_i)$ on countably many inputs $p_0,p_1,p_2,...$ in parallel by a single application of f^∞ as follows: we use the $\langle i,n\rangle$–th application of f to simulate the n–th application of f on input p_i. In this way we can obtain all the results in parallel. This idea can be implemented with the help of the injective recursion theorem. The inverse reduction holds obviously, as $f\leq_{\mathrm{sW}}\widehat{f}$ for every problem f.
(2) We can obtain $f^\diamond(p)$ by f^∞ on a suitable input as follows: we apply f as often as is necessary until we have a successful finite run for $f^\diamond(p)$, then we add "redundant runs" of f on some fixed input from the domain of f^∞ (which is possible as f is pointed). From the result we can read off some value for $f^\diamond(p)$. Again, this idea can be implemented with the help of the injective recursion theorem.
(3) This is just a consequence of (1) and (2). □

The diagram in Figure 3 illustrates the situation for pointed problems.
Another operation related to loops can be defined by $f^\omega:=\mathsf{X}_{n\in\mathbb{N}}f^{[n]}$. For pointed problems we will show that this is equivalent to the parallelization of $f^{[*]}$. For simplicity, we consider this auxiliary operation only for problems on Baire space. Using the realizer version f^{r} of a problem f we can extend everything to arbitrary problems.

Definition 4.3 (Repeated compositional products)**.** We define

(1) $f^{[0]}:=\mathrm{id}$, $f^{[1]}:=\langle\mathrm{id}\times f\rangle$ and
(2) $f^{[n+1]}:=\langle\mathrm{id}\times f\rangle\circ\mathsf{U}\circ f^{[n]}$

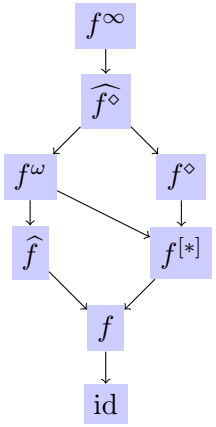


FIGURE 3. Operators on pointed problems.

for all problems $f :\subseteq \mathbb{N}^\mathbb{N} \rightrightarrows \mathbb{N}^\mathbb{N}$ and $n \geq 1$.

Using this definition, we can now define f^ω as follows.

Definition 4.4 (Omega operation)**.** For every problem $f :\subseteq \mathbb{N}^\mathbb{N} \rightrightarrows \mathbb{N}^\mathbb{N}$ we define the problem $f^\omega :\subseteq \mathbb{N}^\mathbb{N} \rightrightarrows \mathbb{N}^\mathbb{N}$ by

$$f^\omega(p) := \langle f^{[0]}(p), f^{[1]}(p), f^{[2]}(p), ...\rangle$$

for all $p \in \mathbb{N}^\mathbb{N}$ such that $p \in \mathrm{dom}(f^{[n]})$ for all $n \in \mathbb{N}$.

Again, this definition can be extended to arbitrary problems f via their realizer version f^r. We note that we could equivalently define

$$f^\omega\langle p_0, p_1, p_2, ...\rangle := \langle f^{[0]}(p_0), f^{[1]}(p_1), f^{[2]}(p_2), ...\rangle.$$

For pointed f this makes no essential difference, as we could always give a dummy input to the first f and use the program input to U to extract components of an input $p = \langle p_0, p_1, p_2, ...\rangle$ step by step as required.

Proposition 4.5 (Omega operation)**.** *For problems $f :\subseteq \mathbb{N}^\mathbb{N} \rightrightarrows \mathbb{N}^\mathbb{N}$ we obtain*

(1) $\widehat{f^\omega} \equiv_\mathrm{sW} f^\omega \leq_\mathrm{sW} \widehat{f^\diamond}$,
(2) $f^{[*]} \leq_\mathrm{sW} f^\omega$, if f is pointed,
(3) $\widehat{f^{[*]}} \equiv_\mathrm{sW} f^\omega$, if f is pointed.

Proof. (1) It is easy to see that $f^{[n]} \leq_\mathrm{sW} f^\diamond$, as we can create a program that runs f exactly n–times with the help of $f^\diamond$ before it comes to a successful halt. This reduction can even be made uniform in n, which allows us to conclude $f^\omega \leq_\mathrm{sW} \widehat{f^\diamond}$. It is also easy to see that f^ω is strongly parallelizable, i.e., $\widehat{f^\omega} \leq_\mathrm{sW} f^\omega$. This is because we have $f^{[n]} \leq_\mathrm{sW} f^{[k]}$ for $n \leq k$ uniformly in n and k.

(2) Since $f^{[n]} \leq_{\mathrm{sW}} f^\omega$ holds for pointed f uniformly in $n \in \mathbb{N}$, we obtain $f^{[*]} \leq_{\mathrm{sW}} f^\omega$.
(3) By (1) this implies $\widehat{f^{[*]}} \leq_{\mathrm{sW}} f^\omega$. The inverse reduction is clear. □

One might be tempted to believe that $f^\omega \equiv_{\mathrm{sW}} \widehat{f^\diamond}$. However, this is only true in certain cases, for instance for single-valued f (see Corollary 4.9) or for problems on Turing degrees (see Proposition 6.1). In order to construct a counterexample for the general case, we use the following lemma. It is well-known that there are strictly descending chains of Turing degrees whose maximum is strictly above the remainder of the sequence.

Lemma 4.6. *There exists a sequence $(p_n)_{n\in\mathbb{N}}$ in $\mathbb{N}^\mathbb{N}$ such that $p_{n+1} <_{\mathrm{T}} p_n$ for all $n \in \mathbb{N}$ and $\langle p_1, p_2, p_3, ...\rangle <_{\mathrm{T}} p_0$.*

For instance, $0 <_{\mathrm{T}} \ldots <_{\mathrm{T}} p_3 <_{\mathrm{T}} p_2 <_{\mathrm{T}} p_1$ could be an initial segment of the Turing degrees [Hug69] and $p_0 := \langle p_1, p_2, p_3, \ldots\rangle'$. We use this fixed sequence for the next example and also for Example 6.2.

Example 4.7. Let $\widehat{0} \in \mathbb{N}^\mathbb{N}$ denote the constant zero sequence. We consider the problem $f :\subseteq \mathbb{N}^\mathbb{N} \rightrightarrows \mathbb{N}^\mathbb{N}$ with $\mathrm{dom}(f) = \{\widehat{0}\} \cup \{p_n : n \in \mathbb{N}\}$ with

$$f(p) := \begin{cases} 0p_0 & \text{if } p = p_0 \\ np_n & \text{if } p = p_{n+1} \\ \{np_n : n \in \mathbb{N}\} & \text{if } p = \widehat{0} \end{cases}$$

Then $f^\diamond \not\leq_{\mathrm{W}} f^\omega$.

Proof. We consider the problem $g : \mathbb{N}^\mathbb{N} \to \mathbb{N}^\mathbb{N}$ with $g(p) = p_0$. We claim that $g \leq_{\mathrm{W}} f^\diamond$. We simply determine $np_n \in f(\widehat{0})$ for some $n \in \mathbb{N}$ and then we need n further applications of f, starting with $f(p_n)$, in order to compute p_0. The first number in the output being 0 indicates success of the computation. On the other hand, $f^\omega(p)$ on some computable input p has a possible output of Turing degree equal to $p_1 <_{\mathrm{T}} p_0$, as $f^{[n]}(p)$ has a possible output kp_k for arbitrary $k \in \mathbb{N}$, hence in particular one of Turing degree equal to p_1 for every $n \in \mathbb{N}$. Thus, $f^\diamond \not\leq_{\mathrm{W}} f^\omega$. □

This example together with the fact that

$$\widehat{\mathsf{C}_\mathbb{N}} \equiv_{\mathrm{sW}} \lim \not\leq_{\mathrm{W}} \mathsf{C}_\mathbb{N} \equiv_{\mathrm{sW}} \mathsf{C}_\mathbb{N}^\diamond \text{ and } \lim^{[2]} \not\leq_{\mathrm{W}} \lim \equiv_{\mathrm{sW}} \widehat{\lim}$$

shows that the diagram in Figure 3 does not allow for any further arrows that involve $f^\diamond$ or $\widehat{f}$.

For single-valued problems $F :\subseteq \mathbb{N}^\mathbb{N} \to \mathbb{N}^\mathbb{N}$ the infinite loop operation coincides with the omega operation. For this result we work out the formal details and we demonstrate again how the injective recursion theorem can be used.

Proposition 4.8 (Infinite loops and the omega operation)**.** *We obtain $F^\omega \equiv_{\mathrm{sW}} F^\infty$ for all single-valued problems $F :\subseteq \mathbb{N}^\mathbb{N} \to \mathbb{N}^\mathbb{N}$.*

Proof. $F^\infty \leq_{\mathrm{sW}} F^\omega$ follows for single-valued F from

$$F^\infty(p) = \langle \mathrm{id} \times \mathsf{U} \times \mathsf{U} \times \mathsf{U} \times ...\rangle \circ F^\omega(p).$$

We still need to prove $F^\omega \leq_{\mathrm{sW}} F^\infty$. For a function $K : \mathbb{N}^\mathbb{N} \to \mathbb{N}^\mathbb{N}$ we write $K_q(p) := K\langle q, p\rangle$ for all $q, p \in \mathbb{N}^\mathbb{N}$. By the injective recursion theorem (Theorem 2.4) there exists a total computable injection $K : \mathbb{N}^\mathbb{N} \to \mathbb{N}^\mathbb{N}$ such that

$$\mathsf{U}_{K_{\langle s,t\rangle}(q)}(r) = \langle K_{\langle q,r\rangle} \times \mathrm{id}\rangle \circ \mathsf{U}\langle q, r\rangle$$

for all $s, t, q, r \in \mathbb{N}^\mathbb{N}$ such that the right-hand side exists. Then we obtain

$$\mathsf{U} \circ \langle \mathrm{id} \times F\rangle \circ \langle K_{\langle s,t\rangle} \times \mathrm{id}\rangle\langle q, p\rangle = \mathsf{U}_{\mathsf{K}_{\langle s,t\rangle}(q)} F(p) = \langle K_{\langle q,F(p)\rangle} \times \mathrm{id}\rangle \circ \mathsf{U} \circ \langle \mathrm{id} \times F\rangle\langle q, p\rangle$$

for all $s,t,q,p\in\mathbb{N}^{\mathbb{N}}$ such that the right-hand side exists. Together with the computable function $K_0:=\langle K_{\widehat{\langle 0,0\rangle}}\times\mathrm{id}\rangle$ we obtain inductively

$$\mathsf{U}\circ F^{[n]}\circ K_0\langle q,p\rangle=\langle K_{F^{[n]}\langle q,p\rangle}\times\mathrm{id}\rangle\circ\mathsf{U}\circ F^{[n]}\langle q,p\rangle$$

for all $q,p\in\mathbb{N}^{\mathbb{N}}$ and $n\geq 1$ such that the right-hand side exists. Let $L,L_0:\mathbb{N}^{\mathbb{N}}\to\mathbb{N}^{\mathbb{N}}$ be computable functions with $L\circ K_{\langle s,t\rangle}(p)=\langle s,t\rangle$ and $L_0\circ K_{\langle s,t\rangle}(p)=p$ for all $s,t,p\in\mathbb{N}^{\mathbb{N}}$ and let $H:\mathbb{N}^{\mathbb{N}}\to\mathbb{N}^{\mathbb{N}}$ be the computable function defined by

$$H\langle\langle p_0,q_0\rangle,\langle p_1,q_1\rangle,\langle p_2,q_2\rangle,...\rangle:=\langle\langle L_0(p_0),q_0\rangle,L(p_1),L(p_2),L(p_3),...\rangle.$$

Now we obtain for suitable $r_n\in\mathbb{N}^{\mathbb{N}}$

$$\begin{aligned}
& H\circ F^{\infty}\circ K_0\langle q,p\rangle\\
= \;& H\circ\langle K_0\langle q,p\rangle,\mathsf{U}\circ F^{[1]}\circ K_0\langle q,p\rangle,\mathsf{U}\circ F^{[2]}\circ K_0\langle q,p\rangle,\mathsf{U}\circ F^{[3]}\circ K_0\langle q,p\rangle,...\rangle\\
= \;& \langle\langle q,p\rangle,L\circ K_{F^{[1]}\langle q,p\rangle}(r_1),L\circ K_{F^{[2]}\langle q,p\rangle}(r_2),L\circ K_{F^{[3]}\langle q,p\rangle}(r_3),...\rangle\\
= \;& \langle F^{[0]},F^{[1]},F^{[2]},F^{[3]},...\rangle\langle q,p\rangle\\
= \;& F^{\omega}\langle q,p\rangle.
\end{aligned}$$

That is $F^{\omega}\leq_{\mathrm{sW}}F^{\infty}$. □

We note that in the special case of pointed f we can conclude $f^{\omega}\leq_{\mathrm{sW}}f^{\infty}$ also from Propositions 4.2 and 4.5. These propositions together with Proposition 4.8 also have the following consequence.

Corollary 4.9 (Infinite loops and single-valuedness)**.** $F^{\infty}\equiv_{\mathrm{sW}}F^{\omega}\equiv_{\mathrm{sW}}\widehat{F^{\diamond}}$ *for single-valued pointed* $F:\subseteq\mathbb{N}^{\mathbb{N}}\to\mathbb{N}^{\mathbb{N}}$.

We use this result in order to determine the infinite loop of LPO and the limit map lim. It turns out that $\lim^{\infty}$ is equivalent to the *ω–Turing jump operator*

$$\mathsf{J}^{(\omega)}:\mathbb{N}^{\mathbb{N}}\to\mathbb{N}^{\mathbb{N}},p\mapsto\langle p,p',p'',...\rangle,$$

which is well-known in computability theory. In the following result we also mention the operator $f\mapsto f^{\dagger}$ that was introduced by Pauly [Pau15] and allows parallelizations and compositions governed by ordinals.

Proposition 4.10 (Infinite loop of LPO and the limit map)**.** *We obtain:*

(1) $\mathsf{LPO}^{\infty}\equiv_{\mathrm{sW}}\mathsf{C}_{\mathbb{N}}^{\infty}\equiv_{\mathrm{sW}}\lim$,

(2) $\lim^{\infty}\equiv_{\mathrm{sW}}\mathsf{J}^{(\omega)}<_{\mathrm{W}}\lim^{\dagger}\equiv_{\mathrm{W}}\mathsf{UC}_{\mathbb{N}^{\mathbb{N}}}$.

Proof. (1) It is clear that

$$\lim\equiv_{\mathrm{sW}}\widehat{\mathsf{LPO}}\leq_{\mathrm{sW}}\mathsf{LPO}^{\infty}\leq_{\mathrm{sW}}\mathsf{C}_{\mathbb{N}}^{\infty}\equiv_{\mathrm{sW}}\lim\nolimits_{\mathbb{N}}^{\infty}$$

holds by Propositions 4.2. We still need to prove $\lim_{\mathbb{N}}^{\infty}\leq_{\mathrm{sW}}\lim$. Since lim is a cylinder, it suffices to show $\lim_{\mathbb{N}}^{\infty}\leq_{\mathrm{W}}\lim$ to complete the proof. To this end we need to show that $\lim_{\mathbb{N}}^{\infty}$ is limit computable. The output $r=\langle\langle q_0,p_0\rangle,\langle q_1,p_1\rangle,\langle q_2,p_2\rangle,...\rangle$ of $\lim_{\mathbb{N}}^{\infty}$ upon input $\langle q_0,p_0\rangle$ can be computed on a limit machine as follows. We inspect the sequence p_0 that is given as input to the first $\lim_{\mathbb{N}}$ and start to compute $\langle q_1,p_1\rangle$ on basis of the assumption that p_0 is constantly $p_0(0)$. As soon as some value $p_1(0)$ is produced, we start computing $\langle q_2,p_2\rangle$ on basis of the assumption that p_1 is constantly $p_1(0)$ and so forth. Simultaneously, we inspect the p_i to check whether the assumptions were justified. As soon as we find a value $p_0(k)\neq p_0(0)$ we restart the entire

process under the assumption that p_0 is constantly $p_0(k)$ from k onwards. After finitely many restarts, caused by p_0, we will not have to change our mind if $p_0 \in \mathrm{dom}(\lim_{\mathbb{N}})$. From this moment on $\langle q_1, p_1\rangle$ will be computed correctly and likewise we will have to change our mind with regards to the assumption on p_1 at most finitely many times. We continue inductively like this in order to compute r and any for any component $\langle q_i, p_i\rangle$ of it only finitely many mind changes are required before the computation eventually produces the correct result.
(2) Firstly, we note that $\mathsf{J}^{(\omega)} \equiv_{\mathrm{sW}} \mathsf{J}^{\omega}$. Here $\mathsf{J}^{(\omega)} \leq_{\mathrm{sW}} \mathsf{J}^{\omega}$ follows, as by the recursion theorem there exists a $q \in \mathbb{N}^{\mathbb{N}}$ with $\mathsf{U}_q(p) = \langle q, p\rangle$ for all $p \in \mathbb{N}^{\mathbb{N}}$. Hence,

$$\mathsf{J}^{[n]}\langle q, p\rangle = \underbrace{\langle \mathrm{id} \times \mathsf{J}\rangle \circ \mathsf{U} \circ \langle \mathrm{id} \times \mathsf{J}\rangle \circ \mathsf{U} \circ .... \circ \mathsf{U} \circ \langle \mathrm{id} \times \mathsf{J}\rangle}_{n\text{–many } \mathsf{J}}\langle q, p\rangle = \langle q, p^{(n)}\rangle$$

for all $p \in \mathbb{N}^{\mathbb{N}}$ and $n \in \mathbb{N}$. The inverse reduction follows as by [Bra18, Theorem 2.14] there is a computable sequence $(H_n)_{n\in\mathbb{N}}$ of computable functions $H_n :\subseteq \mathbb{N}^{\mathbb{N}} \to \mathbb{N}^{\mathbb{N}}$ such that

$$\mathsf{J}^{[n]} = H_n \circ \underbrace{\mathsf{J} \circ ... \circ \mathsf{J}}_{n\text{–times}}.$$

As $\mathsf{J} \equiv_{\mathrm{sW}} \lim$, we obtain $\lim^{\infty} \equiv_{\mathrm{sW}} \mathsf{J}^{(\omega)}$ by Proposition 4.8. By Corollary 4.9

$$\lim{}^{\infty} \equiv_{\mathrm{sW}} \lim{}^{\omega} \leq_{\mathrm{sW}} \lim{}^{\dagger}.$$

The equivalence $\lim^{\dagger} \equiv_{\mathrm{W}} \mathsf{UC}_{\mathbb{N}^{\mathbb{N}}}$ was proved in [Pau15, Theorem 80]. By the same theorem it follows that $\mathsf{J}^{(\omega)} <_{\mathrm{W}} \lim^{\dagger}$, as there are ordinals α with $\omega < \alpha$ such that we obtain $\mathsf{J}^{(\omega)} <_{\mathrm{W}} \mathsf{J}^{(\alpha)} \leq_{\mathrm{W}} \mathsf{J}^{\dagger} \equiv_{\mathrm{sW}} \lim^{\dagger}$, where $\mathsf{J}^{(\alpha)}$ denotes the α–jump operator. □

From this result it follows, in particular, that $f \mapsto f^{\infty}$ is not a closure operator. Secondly, we can conclude that the operators $f \mapsto \widehat{f}$, $f \mapsto f^{\diamond}$, $f \mapsto f^{\infty}$ and $f \mapsto f^{\dagger}$ are pairwise different from each other. In particular, the infinite loop is really a new operation.

Corollary 4.11. *We obtain:*

(1) $\mathsf{LPO}^{\diamond} \equiv_{\mathrm{W}} \mathsf{C}_{\mathbb{N}} <_{\mathrm{W}} \lim \equiv_{\mathrm{sW}} \mathsf{LPO}^{\infty} <_{\mathrm{W}} \mathsf{UC}_{\mathbb{N}^{\mathbb{N}}} \equiv_{\mathrm{W}} \mathsf{LPO}^{\dagger}$,
(2) $\widehat{\lim} \equiv_{\mathrm{W}} \lim <_{\mathrm{W}} \lim^{\diamond} <_{\mathrm{W}} \lim^{\infty}$.
(3) $\mathsf{LPO}^{\infty} \equiv_{\mathrm{sW}} \lim <_{\mathrm{W}} \lim^{\infty} \equiv_{\mathrm{sW}} \mathsf{LPO}^{\infty\infty}$.

Here $\mathsf{UC}_{\mathbb{N}^{\mathbb{N}}} \equiv_{\mathrm{W}} \mathsf{LPO}^{\dagger}$ was proved in [Pau15, Theorem 80] and $\mathsf{LPO}^{\diamond} \equiv_{\mathrm{W}} \mathsf{C}_{\mathbb{N}}$ was proved in [NP18, Proposition 10]. Finally, $\lim^{\infty} \not\leq_{\mathrm{W}} \lim^{\diamond}$ holds, as the former has a non-arithmetical output on some computable inputs, whereas the latter always has arithmetical outputs on computable inputs.

5. Choice and infinite loops

Our next result describes the action of infinite loops with respect to certain closed choice problems. This result can be seen as an infinitary version of the independent choice theorem [BdBP12, Theorem 7.2]. We use the *Sierpiński space* $\mathbb{S} = \{0, 1\}$ with its usual representation $\delta_{\mathbb{S}}$ given by $\delta_{\mathbb{S}}(p) = 0 \iff p = \widehat{0}$.

Theorem 5.1 (Countable independent choice)**.** *Let $A \subseteq \mathbb{N}^{\mathbb{N}}$ and let f be a problem.*

(1) If $f \leq_{\mathrm{W}} \mathsf{C}_A$, then $f^{\infty} \leq_{\mathrm{W}} \mathsf{C}_{A^{\mathbb{N}}}$.
(2) If $f \leq_{\mathrm{W}} \mathsf{UC}_A$, then $f^{\infty} \leq_{\mathrm{W}} \mathsf{UC}_{A^{\mathbb{N}}}$.

Proof. Since $f \equiv_{\mathrm{sW}} f^{\mathrm{r}}$ it is sufficient by Proposition 4.1 to prove the claim for problems $f :\subseteq \mathbb{N}^{\mathbb{N}} \rightrightarrows \mathbb{N}^{\mathbb{N}}$. We prove the first statement. Let $f \leq_{\mathrm{W}} \mathsf{C}_A$. Then f is non-deterministically computable with advice space A according to [BdBP12, Definition 7.1, Theorem 7.2]. That is, there are two computable functions $F_1, F_2 :\subseteq \mathbb{N}^{\mathbb{N}} \to \mathbb{N}^{\mathbb{N}}$ with $\langle \mathrm{dom}(f) \times A\rangle \subseteq \mathrm{dom}(F_2)$ and such that for each $p \in \mathrm{dom}(f)$ we have

(1) $(\exists r \in A)\ \delta_{\mathbb{S}} F_2\langle p, r\rangle = 0$,
(2) $(\forall r \in A)\ (\delta_{\mathbb{S}} F_2\langle p, r\rangle = 0 \Longrightarrow F_1\langle p, r\rangle \in f(p))$.

That is, f can be computed by F_2 with advice $r \in A$, provided the advice is helpful, where F_2 can recognize a non-helpful advice. The first condition guarantees that there is at least one helpful advice.

We need to show that f^∞ is non-deterministically computable with advice space $A^{\mathbb{N}}$. In order to keep track of the individual components, we consider f^∞ with input $\langle q_0, p_0\rangle$, i.e.,

$$f^\infty\langle q_0, p_0\rangle = \{\langle\langle q_0, p_0\rangle, \langle q_1, p_1\rangle, ...\rangle : (\forall i)\ \langle q_{i+1}, p_{i+1}\rangle \in \mathsf{U} \circ \langle \mathrm{id} \times f\rangle\langle q_i, p_i\rangle\}$$

We need to show that there are computable functions $G_1, G_2 :\subseteq \mathbb{N}^{\mathbb{N}} \to \mathbb{N}^{\mathbb{N}}$ with the property $\langle \mathrm{dom}(f^\infty) \times A^{\mathbb{N}}\rangle \subseteq \mathrm{dom}(G_2)$ and such that for each $\langle q_0, p_0\rangle \in \mathrm{dom}(f^\infty)$ we have

(1) $(\exists r \in A^{\mathbb{N}})\ \delta_{\mathbb{S}} G_2\langle\langle q_0, p_0\rangle, r\rangle = 0$,
(2) $(\forall r \in A^{\mathbb{N}})\ (\delta_{\mathbb{S}} G_2\langle\langle q_0, p_0\rangle, r\rangle = 0 \Longrightarrow G_1\langle\langle q_0, p_0\rangle, r\rangle \in f^\infty\langle q_0, p_0\rangle)$.

We define the required computable function G_1 by

$$G_1\langle\langle q_0, p_0\rangle, \langle r_0, r_1, ...\rangle\rangle := \langle\langle q_0, p_0\rangle, \langle q_1, p_1\rangle, ...\rangle$$

for $\langle q_0, p_0\rangle \in \mathbb{N}^{\mathbb{N}}$ and $r_i \in A$ if all the values

$$\langle q_{i+1}, p_{i+1}\rangle := \mathsf{U} \circ \langle \mathrm{id} \times F_1\rangle\langle q_i, \langle p_i, r_i\rangle\rangle$$

exist (and otherwise we leave G_1 undefined). There is also a corresponding computable G_2 that satisfies the following properties for $r := \langle r_0, r_1, r_2, ...\rangle$:

- $(\forall i)(F_2\langle p_i, r_i\rangle = \widehat{0}) \Longrightarrow G_2\langle\langle q_0, p_0\rangle, r\rangle = \widehat{0}$,
- $(\exists k)(F_2\langle p_k, r_k\rangle \in \mathbb{N}^{\mathbb{N}} \setminus \{\widehat{0}\}$ and $(\forall i < k)\ F_2\langle p_i, r_i\rangle = \widehat{0})$
  $\Longrightarrow G_2\langle\langle q_0, p_0\rangle, r\rangle \in \mathbb{N}^{\mathbb{N}} \setminus \{\widehat{0}\}$.

We need to show that G_1 and G_2 satisfy the claims. For this purpose, let $\langle q_0, p_0\rangle \in \mathrm{dom}(f^\infty)$. Then for every $r = \langle r_0, r_1, ...\rangle$ with $r_i \in A$ we have different cases that are to be considered. Firstly, $\langle q_0, p_0\rangle \in \mathrm{dom}(\mathsf{U} \circ \langle \mathrm{id} \times f\rangle)$ and if $\delta_{\mathbb{S}} F_2\langle p_0, r_0\rangle = 0$, then $F_1\langle p_0, r_0\rangle \in f(p_0)$ and hence $\langle q_1, p_1\rangle$ exists and is in $\mathrm{dom}(\mathsf{U} \circ \langle \mathrm{id} \times f\rangle)$. If we continue inductively like this and $\delta_{\mathbb{S}} F_2\langle p_i, r_i\rangle = 0$ for all i, then indeed $\delta_{\mathbb{S}} G_2\langle\langle q_0, p_0\rangle, r\rangle = 0$ and $G_1\langle\langle q_0, p_0\rangle, r\rangle \in f^\infty\langle q_0, p_0\rangle$. Since there is a suitable $r_i \in A$ with $\delta_{\mathbb{S}} F_2\langle p_i, r_i\rangle = 0$ for each i, it follows that, in particular, there is some suitable $r \in A^{\mathbb{N}}$ with $\delta_{\mathbb{S}} G_2\langle\langle q_0, p_0\rangle, r\rangle = 0$ and hence G_2 meets condition (1). Another case for r that we need to consider is the case that eventually there is a first k such that $F_2\langle p_i, r_i\rangle$ still exists but is different from $\widehat{0}$. In this case $G_2\langle\langle q_0, p_0\rangle, r\rangle$ exists and is also different from $\widehat{0}$. Since $\langle \mathrm{dom}(f) \times A\rangle \subseteq \mathrm{dom}(F_2)$, no other case needs to be considered and condition (2) is met too and $\langle \mathrm{dom}(f^\infty) \times A^{\mathbb{N}}\rangle \subseteq \mathrm{dom}(G_2)$.

The statement for $f \leq_{\mathrm{W}} \mathsf{UC}_A$ can be proved analogously, except that a helpful advice $r \in A$ and $r \in A^{\mathbb{N}}$, respectively, is now unique. □

Using $\mathsf{C}_{(A^{\mathbb{N}})^{\mathbb{N}}} \equiv_{\mathrm{sW}} \mathsf{C}_{A^{\mathbb{N}}}$, $\mathsf{UC}_{(A^{\mathbb{N}})^{\mathbb{N}}} \equiv_{\mathrm{sW}} \mathsf{UC}_{A^{\mathbb{N}}}$, we directly obtain the following corollary.

Corollary 5.2. *For every $A \subseteq \mathbb{N}^{\mathbb{N}}$ we obtain:*

(1) $\mathsf{C}_A^\infty \leq_{\mathrm{W}} \mathsf{C}_{A^{\mathbb{N}}}$ *and* $\mathsf{C}_{A^{\mathbb{N}}}^\infty \equiv_{\mathrm{W}} \mathsf{C}_{A^{\mathbb{N}}}$,

(2) $\mathsf{UC}_A^\infty \leq_{\mathrm{W}} \mathsf{UC}_{A^{\mathbb{N}}}$ *and* $\mathsf{UC}_{A^{\mathbb{N}}}^\infty \equiv_{\mathrm{W}} \mathsf{UC}_{A^{\mathbb{N}}}$.

The example $A = \mathbb{N}$ and Proposition 4.10 show that the first reductions cannot be strengthened to equivalences.

Corollary 5.3. $\mathsf{LLPO}^\infty \equiv_{\mathrm{sW}} \mathsf{C}_2^\infty \equiv_{\mathrm{sW}} \mathsf{C}_{2^{\mathbb{N}}} \equiv_{\mathrm{sW}} \mathsf{WKL}$ *and* $\mathsf{C}_{\mathbb{N}}^\infty \equiv_{\mathrm{sW}} \lim <_{\mathrm{W}} \mathsf{C}_{\mathbb{N}^{\mathbb{N}}}$.

We obtain the following interesting fixed points of the infinite loop operation, using Proposition 4.1 and the fact that $\mathsf{C}_{2^{\mathbb{N}}} \equiv_{\mathrm{sW}} \mathsf{WKL}$ and $\mathsf{C}_{\mathbb{N}^{\mathbb{N}}}$ are cylinders.[2]

Corollary 5.4. $\mathsf{C}_{2^{\mathbb{N}}}^\infty \equiv_{\mathrm{sW}} \mathsf{C}_{2^{\mathbb{N}}}$, $\mathsf{C}_{\mathbb{N}^{\mathbb{N}}}^\infty \equiv_{\mathrm{sW}} \mathsf{C}_{\mathbb{N}^{\mathbb{N}}}$, $\mathsf{UC}_{\mathbb{N}^{\mathbb{N}}}^\infty \equiv_{\mathrm{sW}} \mathsf{UC}_{\mathbb{N}^{\mathbb{N}}}$ *and* $\mathsf{WKL}^\infty \equiv_{\mathrm{sW}} \mathsf{WKL}$.

Some classes of problems can be characterized as cones of certain problems, e.g., WKL characterizes the problems that are typically called non-deterministically computable, $\mathsf{C}_{\mathbb{N}}$ characterizes the problems that are computable with finitely many mind changes and lim characterizes the cone of problems that are limit computable (for single-valued problems on computable Polish spaces, $\mathsf{C}_{\mathbb{N}^{\mathbb{N}}}$ characterizes the effectively Borel measurable problems). See [BdBP12] for more details. Hence, the corresponding classes are or are not closed under infinite loops.

Corollary 5.5. *We obtain:*

1. *The class of computable problems, the class of non-deterministically computable problems and the class of single-valued effectively Borel measurable functions (with suitable domains) are closed under infinite loops.*
2. *The class of problems that are computable with finitely many mind changes and the class of problems that are limit computable are not closed under infinite loops.*

6. Loops on computability-theoretic problems

In this section we want to discuss the peculiar situation for problem $f :\subseteq \mathcal{D} \rightrightarrows \mathcal{D}$ on the set $\mathcal{D}$ of *Turing degrees*. We assume that $\mathcal{D}$ is represented by $\deg : \mathbb{N}^{\mathbb{N}} \to \mathcal{D}$, where $\deg(p) := \{q \in \mathbb{N}^{\mathbb{N}} : p \equiv_{\mathrm{T}} q\}$ denotes the Turing degree of $p \in \mathbb{N}^{\mathbb{N}}$. The essential observation is that *while loops* are not more powerful than *for loops* for problems on Turing degrees. The reason is that the stop condition that measures success of a while loop cannot exploit any useful information on Turing degrees (see also the discussion of densely realized problems in [BHK17, BP18]).

Proposition 6.1 (Problems on Turing degrees)**.** *Let $f :\subseteq \mathcal{D} \rightrightarrows \mathcal{D}$ be a problem. Then we obtain*

1. $f^\diamond \equiv_{\mathrm{sW}} f^{[*]}$,
2. $\widehat{f^\diamond} \equiv_{\mathrm{sW}} f^\omega$, *if f is pointed.*

Proof. (1) By Proposition 3.5 we need to prove $f^\diamond \leq_{\mathrm{sW}} f^{[*]}$. Given a name p of an input $\deg(p) \in \mathrm{dom}(f^\diamond)$, there has to be a successful finite run of the loop on f with some finite number $n \in \mathbb{N}$ of loops starting from the name p. During these loops f is applied to data that have been previously computed, i.e., there are $p_0, p_1, ..., p_n \in \mathbb{N}^{\mathbb{N}}$ with $p = p_0$ such that $\deg(p_{i+1}) \in f(\deg(p_i))$ and there is a computable function $g :\subseteq \mathbb{N}^{\mathbb{N}} \to \mathbb{N}$ such that $g\langle p_0, ..., p_n\rangle = 0$ signals success of the finite run. As g needs to be continuous, one can systematically search for $n \in \mathbb{N}$ and $w_0, ..., w_n \in \mathbb{N}^*$ such that the prefixes $w_i \sqsubseteq p_i$ are already sufficient for g to signal success. Now the computation of $f^\diamond$ on the name p can be simulated with the help of $f^{[n]}$ on input p, as the results $\deg(q_{i+1}) \in f(\deg(q_i))$ for $i = 0, ..., n$ and $q_0 = p$ also satisfy $\deg(w_{i+1}q_{i+1}) \in f(\deg(q_i))$ and hence the run of $f^\diamond$ with these modified results $w_{i+1}q_{i+1}$ will also signal success after n loops.

(2) Together with Proposition 4.5 we obtain $\widehat{f^\diamond} \leq_{\mathrm{sW}} \widehat{f^\omega} \equiv_{\mathrm{sW}} f^\omega \leq_{\mathrm{sW}} \widehat{f^\diamond}$ and hence $\widehat{f^\diamond} \equiv_{\mathrm{sW}} f^\omega$ for pointed f. □

[2]The fact that WKL is closed under infinite loops was independently also proved by Pauly et al. (unpublished, personal communication).

However, unlike in the case of single-valued problems, f^∞ remains more powerful than f^ω for problems on Turing degrees in general. We use again the sequence $(p_n)_{n\in\mathbb{N}}$ from Lemma 4.6 and a function similar to the one from Example 4.7 to construct a counterexample. By $a_n := \deg(p_n)$ we denote the Turing degree of p_n for all $n \in \mathbb{N}$.

Example 6.2. We consider $f :\subseteq \mathcal{D} \rightrightarrows \mathcal{D}$ with $\mathrm{dom}(f) = \{0\} \cup \{a_n : n \in \mathbb{N}\}$ with

$$f(a) := \begin{cases} a_0 & \text{if } a = a_0 \\ a_n & \text{if } a = a_{n+1} \\ \{a_n : n \in \mathbb{N}\} & \text{if } a = 0 \end{cases}$$

Then $f^\infty \not\leq_{\mathrm{W}} f^\omega \equiv_{\mathrm{sW}} \widehat{f^\diamond}$.

Proof. We consider the problem $g : \mathcal{D} \to \mathcal{D}$ with $g(a) = a_0$. We claim that $g \leq_{\mathrm{W}} f^\infty$. This is because the output of f^∞ on any input can be used to produce an output of degree a_0. This is because $f(a)$ yields some a_n with $n \in \mathbb{N}$ and n further applications of f yield a_0. On the other hand, $f^\omega(0)$ has a possible output of Turing degree equal to $a_1 < a_0$, as $f^{[n]}(0)$ has a possible output of Turing degree equal to a_1 for every $n \in \mathbb{N}$. Thus, $f^\infty \not\leq_{\mathrm{W}} f^\omega$. Finally, $f^\omega \equiv_{\mathrm{sW}} \widehat{f^\diamond}$ holds by Proposition 6.1. □

It would be nice to have a more natural example of a problem for which f^∞ and $\widehat{f^\diamond}$ are not equivalent. However, our results show that for many natural problems f we actually obtain $f^\infty \equiv_{\mathrm{sW}} \widehat{f^\diamond}$.

Corollary 6.3. *The problems* LPO, LLPO, $\mathsf{C}_{\mathbb{N}}$, $\mathsf{C}_{2^{\mathbb{N}}}$, $\mathsf{C}_{\mathbb{N}^{\mathbb{N}}}$, *and* $\lim$ *in the role of* f *all satisfy* $f^\infty \equiv_{\mathrm{sW}} \widehat{f^\diamond}$.

Analogously, this also holds for many further problems. A natural candidate for a problem that does not satisfy $f^\infty \equiv_{\mathrm{sW}} \widehat{f^\diamond}$ is the *non-computability problem*, which was studied, for instance, in [BHK17, Bra23]:

$$\mathsf{NON} : \mathbb{N}^{\mathbb{N}} \rightrightarrows \mathbb{N}^{\mathbb{N}}, p \mapsto \{q \in \mathbb{N}^{\mathbb{N}} : q \not\leq_{\mathrm{T}} p\}.$$

This problem can also be seen as a problem on Turing degrees. Hence, Proposition 6.1 yields the following.

Corollary 6.4. $\mathsf{NON}^\diamond \equiv_{\mathrm{sW}} \mathsf{NON}^{[*]}$ *and* $\widehat{\mathsf{NON}^\diamond} \equiv_{\mathrm{sW}} \mathsf{NON}^\omega$.

It is easy to see that loops of this problem are related to the following auxiliary problems, which are of independent interest, namely the problems of finding increasing and decreasing chains of Turing degrees above the input:

Definition 6.5 (Increasing and decreasing chains of Turing degrees)**.** We consider the following problems:

(1) $\mathsf{TDINC} : \mathbb{N}^{\mathbb{N}} \rightrightarrows \mathbb{N}^{\mathbb{N}}, p \mapsto \{\langle p_0, p_1, p_2, ...\rangle \in \mathbb{N}^{\mathbb{N}} : (\forall i)\ p <_{\mathrm{T}} p_i <_{\mathrm{T}} p_{i+1}\}$.
(2) $\mathsf{TDDEC} : \mathbb{N}^{\mathbb{N}} \rightrightarrows \mathbb{N}^{\mathbb{N}}, p \mapsto \{\langle p_0, p_1, p_2, ...\rangle \in \mathbb{N}^{\mathbb{N}} : (\forall i)\ p <_{\mathrm{T}} p_{i+1} <_{\mathrm{T}} p_i\}$.

First, we prove that infinite loops of NON are equivalent to TDINC.

Proposition 6.6. $\mathsf{NON}^\infty \equiv_{\mathrm{W}} \mathsf{TDINC}$.

Proof. Firstly, it is clear that $\mathsf{TDINC} \leq_{\mathrm{W}} \mathsf{NON}^{\infty}$: given $p \in \mathbb{N}^{\mathbb{N}}$, we can use NON to compute some q with $q \not\leq_{\mathrm{T}} p$ and hence $p_0 := \langle q, p\rangle$ satisfies $p <_{\mathrm{T}} p_0$. Using NON on input p_0 we obtain q_0 with $q_0 \not\leq_{\mathrm{T}} p_0$ and hence $p_1 := \langle q_0, p_0\rangle$ satisfies $p_0 <_{\mathrm{T}} p_1$. If we continue inductively like this, we obtain an increasing chain $\langle p_0, p_1, p_2, ...\rangle \in \mathsf{TDINC}(p)$.

For the inverse reduction $\mathsf{NON}^{\infty} \leq_{\mathrm{W}} \mathsf{TDINC}$ we consider some input p for NON^{∞} and compute $\langle p_0, p_1, p_2, ...\rangle \in \mathsf{TDINC}(p)$. If in the infinite loop NON^{∞} the problem NON is applied to some input that has been computed from p, then p_0 is a legitimate answer. If in the next loop NON is applied to some result that has been computed from p_0, then p_1 is a legitimate answer and so forth. Hence, the increasing chain $\langle p_0, p_1, p_2, ...\rangle$ can be used to compute answers to all applications of NON in the course of the infinite loop. □

On the other hand, it is easy to see that NON^{ω} can be computed from TDDEC.

Proposition 6.7. $\mathsf{NON}^{\omega} \leq_{\mathrm{W}} \mathsf{TDDEC}$.

Proof. Given an input $p \in \mathbb{N}^{\mathbb{N}}$ to NON^{ω} and a decreasing chain

$$p <_{\mathrm{T}} ... <_{\mathrm{T}} p_2 <_{\mathrm{T}} p_1 <_{\mathrm{T}} p_0$$

of Turing degrees, we can use $\langle p_{n-1}, p_{n-2}, ..., p_0\rangle$ in order to determine an answer for $\mathsf{NON}^{[n]}(p)$ and in this way we can compute a solution in $\mathsf{NON}^{\omega}(p)$. □

We conjecture that NON^{ω} is strictly weaker than NON^{∞}.

Conjecture 6.8. $\mathsf{NON}^{\omega} <_{\mathrm{W}} \mathsf{NON}^{\infty}$.

Given the results of this section, this conjecture is a consequence of the next conjecture.

Conjecture 6.9. $\mathsf{TDINC} \not\leq_{\mathrm{W}} \mathsf{TDDEC}$.

One might be tempted to prove this separation by considering an initial segment of Turing degrees similar as in Lemma 4.6. However, such an initial segment cannot be uniformly computable in the maximal element by results in [Ish02]. Hence, additional ideas are required to resolve the two conjectures stated here.[3] However, one can use initial segments in order to prove the following result.

Proposition 6.10. $\mathsf{NON}^{[n+1]} \not\leq_{\mathrm{W}} \mathsf{NON}^{[n]}$ *for all* $n \in \mathbb{N}$.

Proof. Similar as in Proposition 6.6 one can prove that the problem $f_n : \mathcal{D} \rightrightarrows \mathcal{D}^n$ with

$$f_n(a) := \{(a_1, ..., a_n) \in \mathcal{D}^n : a < a_1 < a_2 < ... < a_n\}$$

satisfies $\mathsf{NON}^{[n]} \equiv_{\mathrm{W}} f_n$ for all $n \in \mathbb{N}$. Now, upon input of 0, the problem f_n can yield an output $(a_1, ..., a_n)$ for an initial segment

$$0 < a_1 < a_2 < ... < a_n$$

of the Turing degrees. From $(a_1, ..., a_n)$ no other Turing degrees outside of this initial segment are computable, in particular, no chain of Turing degrees of length $n + 1$ above 0 can be computed. □

[3] According to the author's understanding, Patrick Uftring has recently solved these conjectures in the affirmative (personal communication).

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Acknowledgments

The author was funded by the German Research Foundation (DFG, Deutsche Forschungsgemeinschaft) – project number 554999067 and by the National Research Foundation of South Africa (NRF) – grant number 151597. We would like to thank Arno Pauly, Manlio Valenti and Hendrik Smischliaew for helpful discussions.